§7-4 Integration of Rational Function by Partial Fractions

Homework：9,11,23,26,29,31,37,39,45,47

How to integrate $\int \frac{P(x)}{Q(x)}\,dx$？Here $P(x)$，$Q(x)$ are polynomials？

Step1：Write $\frac{P(x)}{Q(x)}$ as $\frac{P(x)}{Q(x)} = S(x) + \frac{R(x)}{Q(x)}$,where $\deg R < \deg Q$.

Step2：將$Q(x)$因式分解成一次或二次的連乘積,即$Q(x)$的因式基本型為

$(ax^2+bx+c)^n$, 和$(ex+f)^m$, where $b^2-4ac<0$, and n and m are nonnegative integers.

Step3：

i. 含$(ex+f)^m$的因式，將$\frac{R(x)}{Q(x)}$寫成 partial fractions（部份分式）時，需含有下列各項

$$(ex+f)^m \to \frac{f_1}{ex+f} + \frac{f_2}{(ex+f)^2} + \ldots\ldots + \frac{f_m}{(ex+f)^m} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \quad (1)$$

ii. 同理含 $(ax^2+bx+c)^n$ 的因式，部份分式需含下列各項

$$(ax^2+bx+c)^n \to \frac{b_1x+c_1}{(ax^2+bx+c)} + \frac{b_2x+c_2}{(ax^2+bx+c)^2} + \ldots\ldots + \frac{b_nx+c_n}{(ax^2+bx+c)^n} \ldots\ldots(2)$$

Why？

給一$(m-1)$次多項式$R(x)$,可將$R(x)$唯一的表成（Why？）

$$R(x) = f_1(ex+f)^{m-1} + f_2(ex+f)^{m-2} + \ldots\ldots + f_{m-1}(ex+f) + f_m$$

（即將x轉成$ex+f$）

$$\Rightarrow \frac{R(x)}{(ax+b)^m} = \frac{f_1}{ex+f} + \frac{f_2}{(ex+f)^2} + \ldots\ldots + \frac{f_{m-1}}{(ex+f)^{m-1}} + \frac{f_m}{(ex+f)^m}.$$

Step4：

i. (1)式每一項,都很容易逐項積分.

ii. (2)式每一項,都可透過配方法及 substitution 寫成

$$\frac{\alpha u+\beta}{\left(u^2+r\right)^i}=\frac{\alpha u}{\left(u^2+r\right)^i}+\frac{\beta}{\left(u^2+r\right)^i}.$$

上式的第一項可用 direct substitution.

上次的第二項可用三角 substitution.

Example 1： $\int \frac{x^4-2x^2+4x+1}{x^3-x^2-x+1}.$

Solution：

$$\frac{x^4-2x^2+4x+1}{x^3-x^2-x+1}\overset{\text{長除法}}{=\!=\!=}(x+1)+\frac{4x}{(x+1)(x-1)^2}$$

$$\frac{R(x)}{Q(x)}=\frac{4x}{(x+1)(x-1)^2}=\frac{a}{x+1}+\frac{b}{(x-1)}+\frac{c}{(x-1)^2}$$

$$\Rightarrow \quad 4x=a(x-1)^2+b(x+1)(x-1)+c(x+1)$$

$$x=-1 \quad \Rightarrow \quad a=-1$$

$$x=1 \quad \Rightarrow \quad c=2$$

比較 2 次項的係項：$0=a+b\Rightarrow b=1$

$$\Rightarrow \text{原式} =\int (x+1)dx+\int\left(\frac{-1}{x+1}\right)+\frac{1}{x-1}+\frac{2}{(x-1)^2}dx$$

$$=\frac{x^2}{2}+x-\ln|x+1|+\ln|x-1|-\frac{2}{(x-1)}+\text{C}.$$

Example 2： $\int \frac{2x^2-x+4}{x^3+4x}dx.$

Solution：

$$\frac{2x^2-x+4}{x^3+4x}=\frac{2x^2-x+4}{x\left(x^2+4\right)}=\frac{a}{x}+\frac{bx+c}{x^2+4}$$

$$2x^2-x+4=a\left(x^2+4\right)+x(bx+c)$$

$$x=0\Rightarrow a=1$$

比較係數$\Rightarrow\begin{cases} a+b=2 & \Rightarrow \quad b=1 \\ c=-1 \end{cases}$

$$\Rightarrow \text{原式} = \int \frac{1}{x}+\frac{x-1}{x^2+4}dx$$

$$= \int\left(\frac{1}{x}+\frac{x}{x^2+4}-\frac{1}{x^2+4}\right)dx$$

$$= \ln|x|+\frac{1}{2}\ln\left(x^2+4\right)-\frac{1}{2}\tan^{-1}\left(\frac{x}{2}\right)+\text{C}.$$

Example 3： $\int\frac{4x^2-3x+2}{4x^2-4x+3}dx$.

Solution：

$$\int\frac{4x^2-3x+2}{4x^2-4x+3}dx$$

$$= \int 1dx+\int\frac{x+1}{(2x-1)^2+2}dx \qquad \begin{pmatrix} u=\text{-}1\Rightarrow du=2dx \\ x=\dfrac{u+1}{2} \end{pmatrix}$$

$$= x+\frac{1}{2}\int\frac{\frac{u+1}{2}-1}{u^2+2}du$$

$$= x+\frac{1}{4}\int\frac{u-1}{u^2+2}du$$

$$= x+\frac{1}{8}\ln\left(u^2+2\right)-\frac{1}{4\sqrt{2}}\tan^{-1}\frac{x}{\sqrt{2}}+\text{C}.$$

Example 4： $\int\frac{1-x+2x^2-x^3}{x\left(x^2+1\right)^2}dx.$

Solution：

$$\frac{1-x+2x^2-x^3}{x\left(x^2+1\right)^2}=\frac{a}{x}+\frac{bx+c}{x^2+1}+\frac{dx+e}{\left(x^2+1\right)^2}$$

$$\Rightarrow\left(1-x+2x^2-x^3\right)=a\left(x^2+1\right)^2+x\left(bx+c\right)\left(x^2+1\right)+x\left(dx+e\right)$$

$$x=0 \quad\Rightarrow\quad a=1$$

x^4 的係數 $\quad 0 = a + b \quad \Rightarrow b = -1$

x^3 的係數 $\quad -1 = c$

x 的係數 $\quad -1 = c + f \quad \Rightarrow e = 0$

x^2 的係數 $\quad 2 = 2a + b + d \quad \Rightarrow d = 1$

原式 $= \ln|x| - \frac{1}{2}\ln\left(x^2+1\right) - \tan^{-1}x - \frac{1}{2\left(x^2+1\right)} + \text{C}.$

Example 5： $\int \frac{1}{x\sqrt{x+1}}dx..$

Solution：

$u = (x+1)^{\frac{1}{2}} \quad u^2 = x+1 \quad \Rightarrow \quad 2udu = dx$

$$\int \frac{1}{x\sqrt{x+1}}dx$$

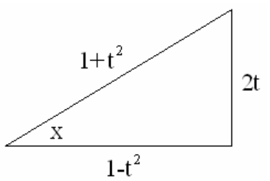
$$= \int \frac{2udu}{u\left(u^2-1\right)}.$$

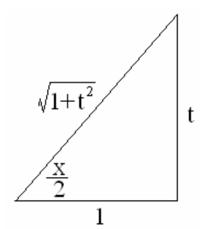
$$= \int \left(\frac{1}{u-1} - \frac{1}{u+1}\right)du$$

$$= \ln\left|\frac{\sqrt{x+1}-1}{\sqrt{x+1}+1}\right| + \text{C}.$$

Example 6： $\int \frac{dx}{3\sin - 4\cos x}.$

Solution：





$t = \tan\frac{x}{2} \Rightarrow \tan x = \tan\left(\frac{x}{2}+\frac{x}{2}\right) = \frac{2t}{1-t^2}$

$\Rightarrow \sin x = \frac{2t}{1+t^2}, \quad \cos x = \frac{1-t^2}{1+t^2}$

$dt = \frac{1}{2}\sec^2\frac{x}{2}dx = \frac{1}{2}\left(1+t^2\right)dx$

$\Rightarrow dx = \frac{2}{1+t^2}dt$

原式 $= \int \frac{P(t)}{Q(t)}dt = \int \frac{1}{2t^2+3t-2}dt$

即任一 rational function of $\sin x$ and $\cos x$

可改成 rational function of $t, t = \tan\frac{x}{2}.$